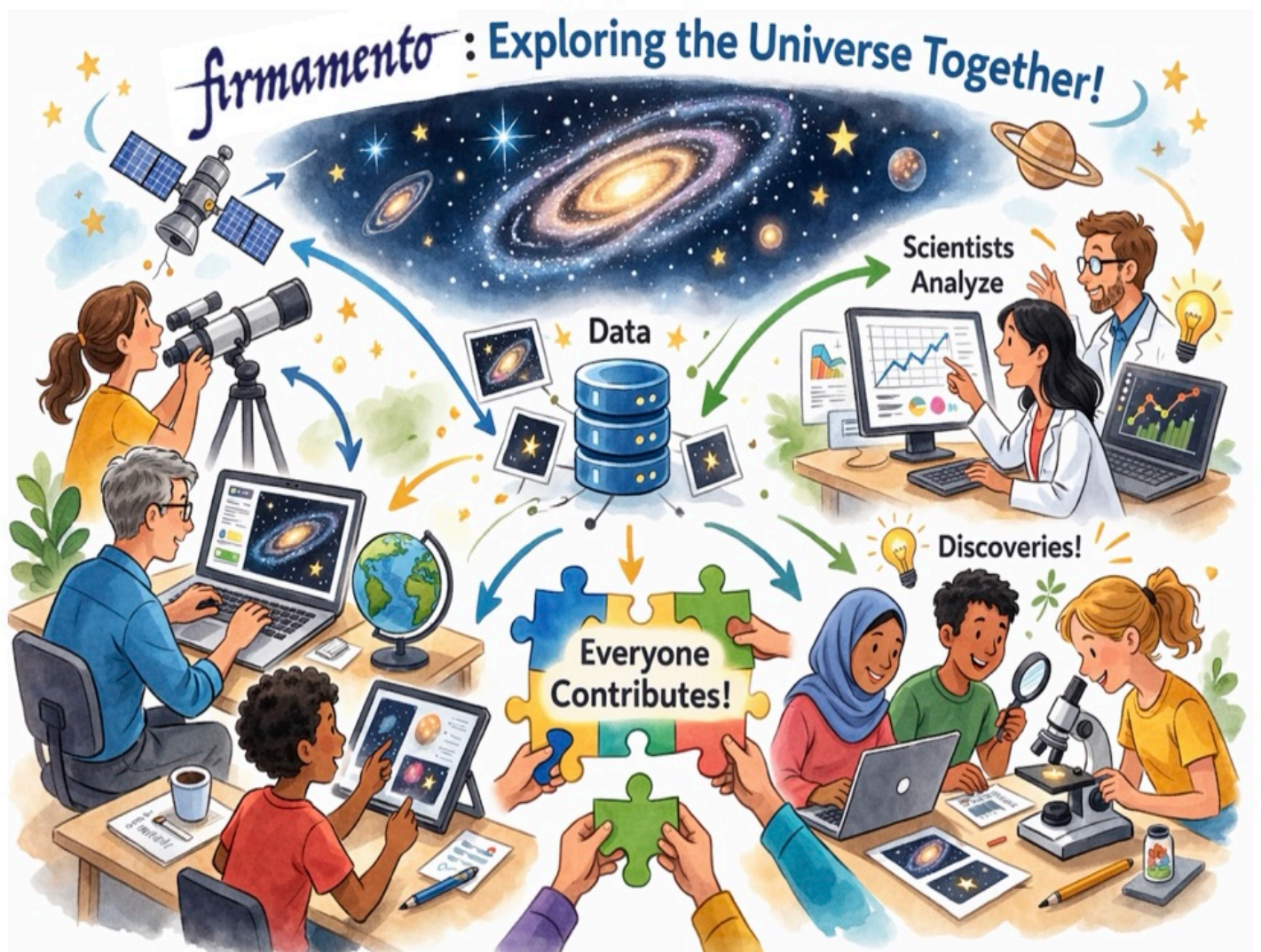


Public Participation in Astronomy with Firmamento

June 20, 2026



Public Participation in Astronomical Research with Firmamento

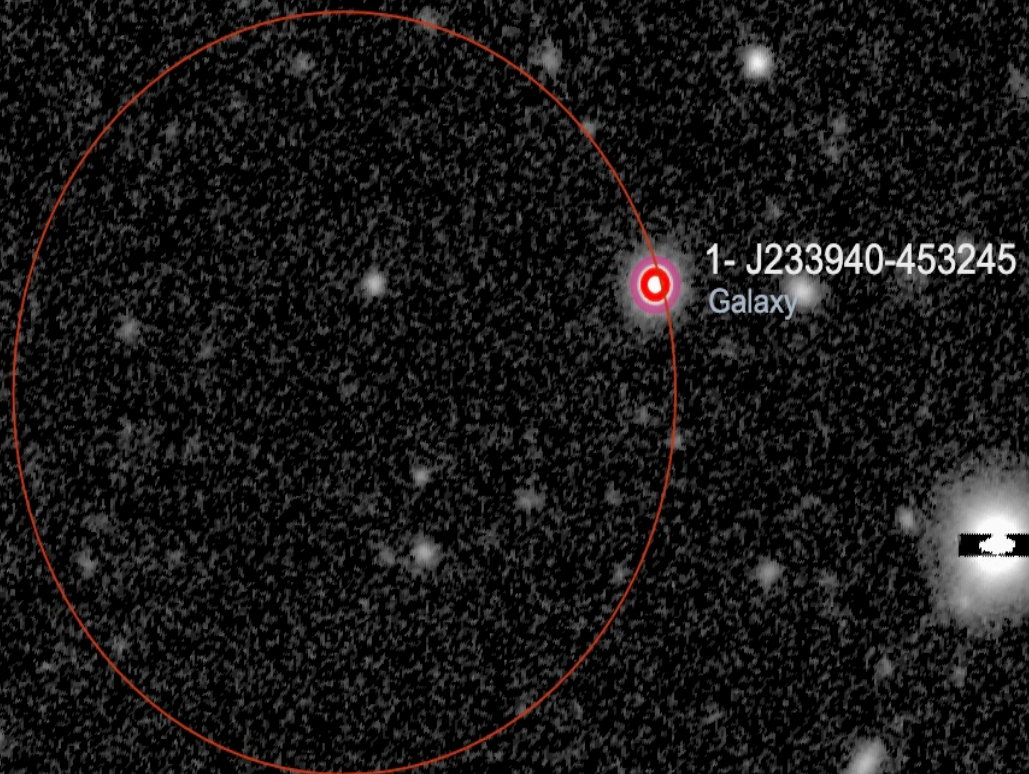
The vision of making science-level data accessible to everyone, once seen as nearly utopian, is now becoming a tangible reality. This transformation is driven by several key factors, including: (a) the widespread availability of the internet; (b) the adoption of open-access policies for many satellite and ground-based datasets; (c) the availability of a new generation of tools that provide easy access to ready-to-use datasets; and (d) the rapid advancements in artificial intelligence and machine-learning techniques.

Firmamento and Fbook are platforms built to take advantage of the current favorable conditions and offer opportunities for non-experts to participate in high-quality scientific research. Both tools embrace the principles of the United Nations **Open Universe Initiative** and seek to be inclusive, fostering participation in scientific discovery by all, in line with the Open Universe's long-term goal of building capacity and democratizing science.

Citizen science - a well-established practice in which non-experts contribute to research projects conceived and designed by professional scientists - together with the emerging concept of citizen-driven exploration, offers powerful ways to involve non-professional participants in astronomical research.

The next sections introduce some initial examples of contributions to scientific research by non-experts using the tools of Firmamento and Fbook. More projects will be proposed in future versions of this eBook.

0.1 Shooting in the (cosmic) dark: discover stars, galaxies, quasars, blazars... and perhaps more



Discover the variety of astronomical sources that populate the sky.

There are many trillions of stars in the Universe and probably one trillion galaxies, together with billions of black holes, active galactic nuclei (AGN), and other exotic objects. Most of them are simply anonymous points of light in the sky and do not appear in any catalog of known sources.

Firmamento provides a tool that, given a position in the sky, locates the sources in that region and identify their nature if they are sufficiently bright and multi-frequency data are available in open archives. That is, Firmamento can indicate whether a source is likely to be a star, a galaxy, a QSO, a blazar, etc., or possibly something completely new.

This capability can be used both to explore the sky and to study it in qualitative and quantitative ways. For instance, by pointing at random positions away from the Galactic plane to avoid crowded regions, Firmamento enables an **unbiased** exploration that reveals in a completely objective and quantitative way the types of sources present, provides information on their surface density, and may lead to the discovery of new objects belonging to known categories, or even entirely new classes of sources with peculiar multi-frequency emission.

Click the icon below to **point to a random position in the sky** and explore the region, identifying local sources.



If Firmamento finds one or more sources within the (one-square-arcminute¹) search area, it provides the name of the constellation, the precise position(s), the identifier(s) of the source(s) in the official **IAU nomenclature**, the source(s) types, and multi-frequency data.

You will probably notice that most of the sources found by pointing completely randomly in the sky are stars and some galaxies. To find rare sources such as radio galaxies, QSOs, and blazars, it is more efficient to point toward the direction of a known radio or X-ray source. This approach is discussed in the next section.

¹One square arcminute of sky is a tiny area! For comparison, the full moon, on average, covers an area of more than 700 square arcminutes.

0.2 Pointing at X-ray sources: discover AGN, radio galaxies, quasars, blazars... and more



Discover radio galaxies, quasars and blazars among X-ray or radio emitting astronomical sources

In this section the Firmamento tool described above - which identifies the nature of sources on the basis of multi-frequency data - can be run on X-ray sources that are spatially coincident with known radio sources and are located away from the Galactic plane. This type of objects may be radio galaxies, Seyfert galaxies, pulsars, quasars, blazars or special types of stars or binary systems. Other source types are rarely found.

Click the icons below (from left to right) to point at

- randomly chosen Swift X-ray serendipitous sources that are coincident with radio sources.
- randomly chosen X-ray sources that also show radio emission and are located in the eROSITA-eFEDS region
- randomly chosen X-ray sources in the eRASS1 survey that are not detected in radio surveys
- randomly chosen sources from the Radio Fundamental Catalog (RFC) with radio flux density larger than 50 mJy

Swift-
XRT+radio



eFEDS + radio



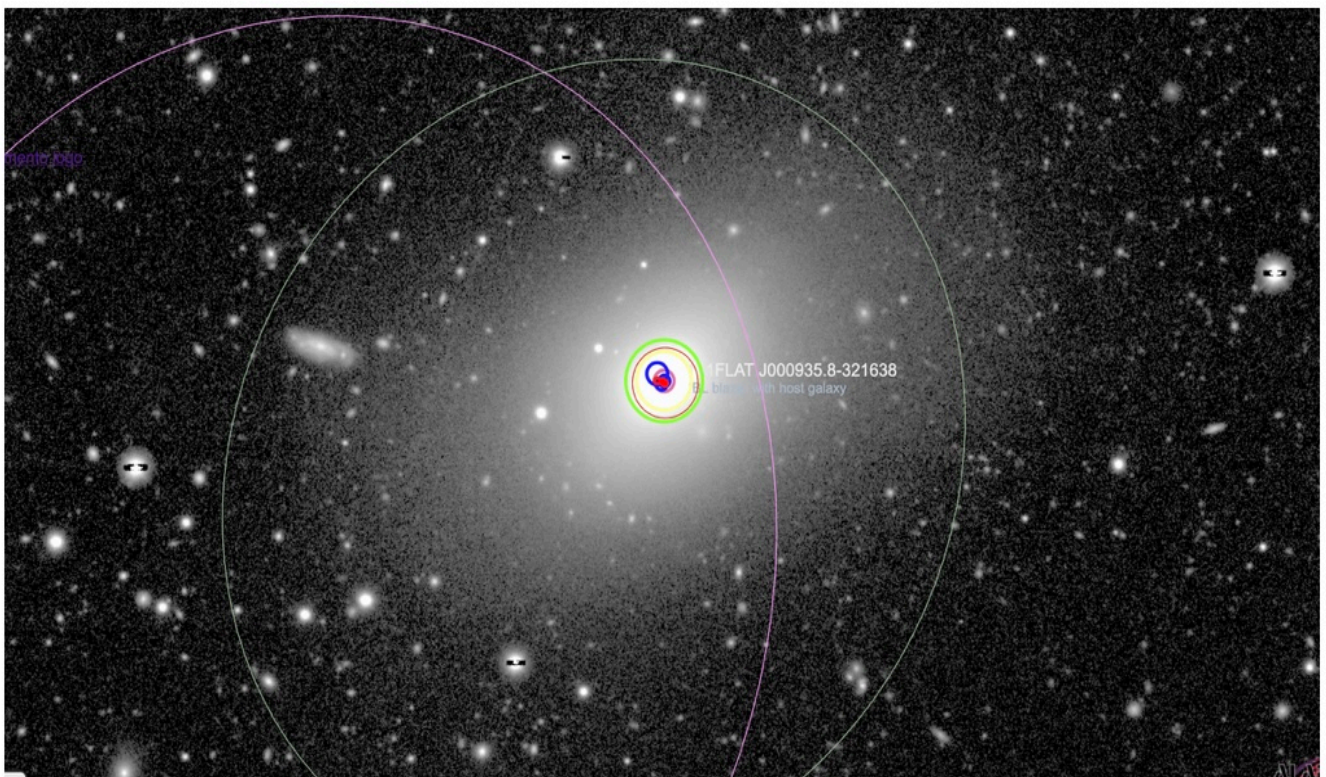
eROSITA



Radio from RFC



0.3 Help discover the counterparts of the γ -ray sources in the latest Fermi-LAT catalog: FL16Y



Participate in the discovery of the counterpart of unassociated Fermi-LAT γ -ray sources

One interesting and very timely citizen science project within Firmamento and Fbook involves the participation of non-experts in the identification of counterparts of γ -ray sources listed in the newest Fermi-LAT γ -ray source catalog: FL16Y. This catalog is based on 16 years of observations with the Large Area Telescope (LAT) on board the NASA Fermi satellite (Ballet et al. 2026).

What is the counterpart of a γ -ray source?

An astrophysical gamma-ray source is often detected with a large location uncertainty because γ -rays are difficult to pinpoint precisely. Its "counterpart" is the object or source seen in other types of light, such as visible, X-ray, or radio waves that matches or explains the γ -ray emission. The counterpart of a γ -ray source is therefore the same physical object, located within or very close to the uncertainty region.

The FL16Y catalog is reproduced in the **Google Sheet** shown in Fig.1 below), where each source is listed together with related information from previous catalogs. Each source is identified by a name such as FL16Y J0006.75244, where the numerical part encodes its position in the sky. For each entry, the results of the 4FGL-DR4 catalog and the corresponding counterparts from the 4FGL, 4LAC, and 1FLAT catalogs - derived from earlier analyses - are also provided.

	A	G	M	P	Q	R
1	FL16Y_Name	4FGL_Name	1FLAT_ASSOC	4FGL_ASSOC1	4LAC_ASSOC1	16Y_ASSOC1
2	FL16Y J0000.0+3145	4FGL J2359.9+3145	1FLAT J235955.3+314600		NVSS J235955+314558	NVSS J235955+314558
3	FL16Y J0000.4-7354	4FGL J0000.3-7355				
4	FL16Y J0000.5+0747	4FGL J0000.5+0743				
5	FL16Y J0000.9+2531	4FGL J0000.7+2530	1FLAT J000027.9+252804			
6	FL16Y J0001.2-0747	4FGL J0001.2-0747	1FLAT J000118.0-074626	PMN J0001-0746	PMN J0001-0746	PMN J0001-0746
7	FL16Y J0001.2+4741	4FGL J0001.2+4741	1FLAT J000119.0+474200	B3 2358+474	B3 2358+474	B3 2358+474
8	FL16Y J0001.5-4155	4FGL J0001.6-4156	1FLAT J000132.7-415524	2MASS J00013275-415525	2MASS J00013275-415525	2MASS J00013275-415525
9	FL16Y J0001.5-0010	4FGL J0001.4-0010	1FLAT J000121.5-001139	FBQS J0001-0011	FBQS J0001-0011	FBQS J0001-0011
10	FL16Y J0001.6+2113	4FGL J0001.5+2113	1FLAT J000132.3+211336	TXS 2358+209	TXS 2358+209	TXS 2358+209
11	FL16Y J0001.6+3501	4FGL J0001.6+3503				
12	FL16Y J0002.2+6719					1RXS J000209.4+672503
13	FL16Y J0002.3-6728	4FGL J0002.1-6728	1FLAT J000215.4-672652	SUMSS J000215-672653	SUMSS J000215-672653	SUMSS J000215-672653
14	FL16Y J0002.4-0815	4FGL J0002.3-0815	1FLAT J000236.1-081532	WISEA J000236.06-081532	WISEA J000236.06-081532.4	WISEA J000236.06-081532.4
15	FL16Y J0002.4-5156	4FGL J0002.4-5156	1FLAT J000229.2-515227	WISE J000229.20-515227.4	WISE J000229.20-515227.4	WISE J000229.20-515227.4

Figure 1: The first rows of the Google Sheet containing the FL6Y catalog

Within the Google sheet, clicking on a source name (FL16Y or 4FGL) opens the corresponding Firmamento page, where detailed information about the source is displayed together with its positional uncertainty region (usually an ellipse) overlaid on an optical survey image. Clicking on a 1FLAT name instead provides access to the Firmamento page containing details of the corresponding 1FLAT counterpart, including its spectral energy distribution (SED) and additional data. Once in Firmamento, γ -ray source error ellipses, candidate counterparts, detailed SEDs, and additional information can be analyzed interactively.

Step-by-step instructions on how to work with the catalog, display data and find possible γ -ray sources counterparts, as well as video tutorials are available in **Fbook**. Click on "Your Universe - Citizen science and education" button, then on

“citizen science” and finally on “Identification of γ -ray sources in the 4FGL-DR4 catalog” as shown in figure 2.

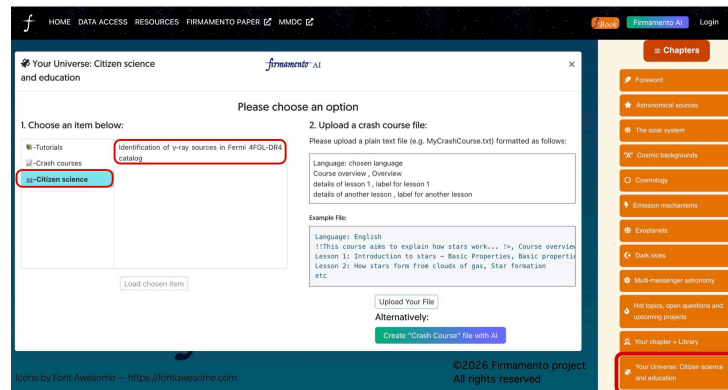


Figure 2: Figure illustrating how to select the Citizen science project titled: Identification of γ -ray sources in Fermi 4FGL-DR4 catalog

After loading the project, to step-by-step instructions can be accessed as shown in Fig.3.

Figure 3: Fbook dropdown menu showing the links to the step-by-step procedure and related information.

More details and a **specific video tutorial (in preparation)** will be provided soon.